

Shobhita Sundaram

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ssundaram21.github.io

EDUCATION

Massachusetts Institute of Technology (MIT)	Cambridge, MA
Ph.D. Computer Science	2022–Present
S.M. Computer Science	2022–2025
Advisor: Phillip Isola	
Massachusetts Institute of Technology (MIT)	Cambridge, MA
S.B. Computer Science, S.B. Mathematics	2018–2022
Advisors: Pawan Sinha, Xavier Boix, Tomaso Poggio	

PUBLICATIONS

* indicates equal contribution

1. Teaching Models to Teach Themselves: Reasoning at the Edge of Learnability.
S. Sundaram, J. Quan, A. Kwiatkowski, K. Ahuja, Y. Ollivier, J. Kempe. *ICML*, 2026 (**spotlight**).
2. Better Together: Leveraging Unpaired Multimodal Data for Stronger Unimodal Models.
S. Gupta, **S. Sundaram**, C. Wang, S. Jegelka, P. Isola. *ICLR*, 2026.
3. What Makes for a Good Stereoscopic Image?
N. Tamir, S. Amir, R. Itzhaky, N. Atia, **S. Sundaram**, S. Fu, R. Sokolovsky, P. Isola, T. Dekel, R. Zhang, M. Farber. *CVPR Computer Vision for Metaverse Workshop*, 2025.
4. Personalized Representation from Personalized Generation.
S. Sundaram*, J. Chae*, Y. Tian, S. Beery, P. Isola. *ICLR*, 2025.
5. When Does Perceptual Alignment Benefit Vision Representations?
S. Sundaram*, S. Fu*, L. Muttenthaler, N. Tamir, L. Chai, S. Kornblith, T. Darrell, P. Isola. *NeurIPS*, 2024.
6. DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data.
S. Fu*, N. Tamir*, **S. Sundaram***, L. Chai, R. Zhang, T. Dekel, P. Isola. *NeurIPS*, 2023 (**spotlight**).
7. Recurrent Connections Facilitate Symmetry Perception in Deep Networks.
S. Sundaram*, D. Sinha*, M. Groth, T. Sasaki, X. Boix. *Scientific Reports*, 2022.
8. GAN-Based Data Augmentation for Chest X-ray Classification.
S. Sundaram* and N. Hulkund*. *KDD Workshop on Applied Data Science for Healthcare*, 2021.
9. Do Neural Networks for Segmentation Understand Insideness?
K. Villalobos*, V. Štih*, A. Ahmadinejad*, **S. Sundaram**, J. Dozier, A. Franci, F. Azevdo, T. Sasaki, X. Boix. *Neural Computation*, 2021.

EXPERIENCE

FAIR	Paris, France
Research Scientist Intern	May - October 2025
– <i>Project</i> : Asymmetric self-play with LLMs for difficult reasoning problems.	
– <i>Mentors</i> : Julia Kempe, Yann Olivier, Kartik Ahuja	
Google Research	Cambridge, MA
Student Researcher	December 2023 - March 2024

- *Project:* Synthetic data generation with diffusion models for personalizing vision backbones.
- *Mentors:* Yonglong Tian, Dilip Krishnan

Google DeepMind

London, UK

Research Engineering Intern

June - August 2022

- *Project:* Data selection for LLM pretraining.
- *Mentors:* Sebastian Borgeaud, Laurent Sifre, Jordan Hoffman, Arthur Mensch

Center for Brains, Minds, and Machines, MIT

Cambridge, MA

Undergraduate Researcher

September 2019 - May 2022

- *Project:* Recurrent vision models for visual long-range spatial dependencies.
- *Mentors:* Xavier Boix, Pawan Sinha, Tomaso Poggio

The D. E. Shaw Group

New York, NY

Quantitative Research Intern

June - August 2021

- *Project:* RL tools for portfolio management; outperformed optimal control theory baselines.
- *Mentor:* Konstantin Turitsyn

Apple

Cupertino, CA

Machine Learning Intern

June - August 2020

Two Sigma Investments

Houston, TX

Software Engineering Intern

May - August 2019

AWARDS

NSF Graduate Research Fellowship	2022 - 2025
HDTV Grand Alliance Fellowship	2022 - 2023
MIT Undergraduate Research and Innovation Scholar	2020
MIT Burchard Scholar	2020

SERVICE & LEADERSHIP

Organizer: CVPR Workshop on Synthetic Data for Computer Vision	2024, 2025, 2026
Organizer: ECCV Tutorial on Efficient Text-to-Image Modeling	2024
Mentor: MIT Graduate Application Assistant Program	2022 - 2024
Event Coordinator: MIT Graduate Women of EECS	2023
Associate Editor: MIT Science Policy Review	2020 - 2022

INVITED TALKS

Self-Improving at the Frontier of Learnability.	
Lila Sciences	April 2026
The Vector Institute	February 2026
Personalized Representation from Personalized Generation.	
CVPR Syntagen Workshop	June 2025
Cohere for AI	March 2025
Representation Learning with Perceptual Alignment.	
Stanford NeuroAILab	April 2025
Evaluating Text-to-Image Models.	
ECCV Efficient Text-to-Image Modeling Tutorial	September 2024
DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data.	
Adobe	October 2023
Computer Vision Meetup, hosted by Vox51	July 2023